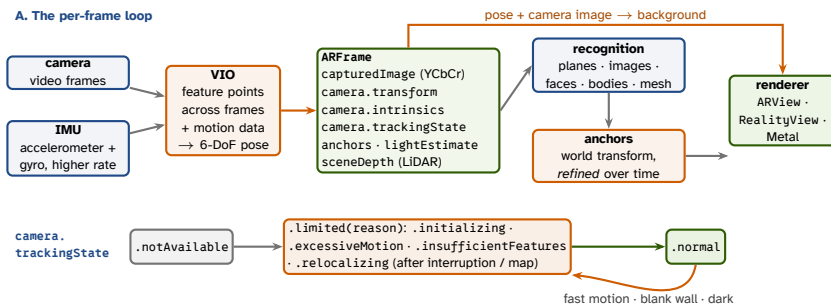
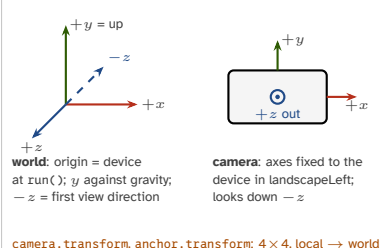


**In one line:** ARKit fuses feature points tracked across camera frames with the **IMU (visual-inertial odometry)** into a 6-DoF **camera pose** in a gravity-aligned world in **metres**; each tick is an **ARFrame**, and what it recognises (planes, images, faces, meshes) becomes **anchors** it keeps correcting. **RealityKit** renders on top: an **Entity-Component-System** engine (ARView in UIKit, RealityView in SwiftUI from iOS 18) drawing USD content locked to those anchors.

A. The per-frame loop



B. Two spaces (right-handed, metres)



## How ARKit works

- ARSession runs **one** configuration; run(cfg, options: [.resetTracking, .removeExistingAnchors]) starts over, pause() frees the camera. The delegate gets frames, anchor add/update/remove, tracking-state changes, interruptions.

AR\_Configuration

|       |                                                                                                                       |
|-------|-----------------------------------------------------------------------------------------------------------------------|
| World | back camera, 6 DoF: planes, images, objects, mesh, depth, people                                                      |
| Face  | front: ARFaceAnchor mesh + blendShapes 0-1 (jawOpen...); TrueDepth up to iOS 13, any Neural Engine device from iOS 14 |
| Body  | ARBodyAnchor skeleton · <b>Image</b> : known images only                                                              |
| Geo   | ARGeoAnchor at lat/long; A12+, GPS, outdoors, supported cities                                                        |

- Anchors:** ARPlaneAnchor (horizontal/vertical, classification) · ARImageAnchor (reference image + **physical size**) · ARObjectAnchor (.arobject) · ARMeshAnchor (LiDAR, sceneReconstruction = .mesh, iOS 13.4).
- Raycast** (iOS 13; hitTest deprecated in 14): screen point → ARRaycastQuery; target .estimatedPlane (instant, rough) · .existingPlaneGeometry (inside a detected plane) · .existingPlaneInfinite. trackedRaycast re-fires as understanding improves; stopTracking().
- Depth:** frameSemantics .personSegmentationWithDepth (iOS 13, people occlude content) · .sceneDepth (iOS 14, LiDAR depth + confidence). Check supportsFrameSemantics first.
- Persist:** at worldMappingStatus == .mapped, getCurrentWorldMap → ARWorldMap (NSData) → later initialWorldMap; state is .relocalizing until the room is recognised. **Shared:** isCollaborationEnabled; you ship CollaborationData (e.g. Multipeer) → update(with:); best ≤ 4 users.
- Limited tracking: show a hint, place nothing; or ARCoachingOverlayView (iOS 13, goal: .horizontalPlane...). lightEstimate: 1000 lm = neutral.

## Example — tap to place (ARView)

```

let cfg = ARWorldTrackingConfiguration()
cfg.planeDetection = [.horizontal]
if ARWorldTrackingConfiguration.supportsSceneReconstruction(.mesh) {
    cfg.sceneReconstruction = .mesh // LiDAR only
}
arView.session.run(cfg)
guard let q = arView.makeRaycastQuery(from: tapPoint,
    allowing: .estimatedPlane, alignment: .horizontal),
    let hit = arView.session.raycast(q).first else { return }
let a = AnchorEntity(raycastResult: hit)
a.addChild(ModelEntity(mesh: .generateBox(size: 0.1), // 10 cm
    materials: [SimpleMaterial()]))
arView.scene.addAnchor(a)
var t = matrix_identity_float4x4; t.columns.3.z = -0.5 // 50 cm ahead
session.add(anchor: ARAnchor(transform: frame.camera.transform * t))
  
```

## RealityKit

- ECS:** Entity = identity + hierarchy; **components** = data (Transform, ModelComponent, CollisionComponent,

PhysicsBodyComponent, AnchoringComponent, yours); **systems** (iOS 15) = behaviour, run each frame over the entities that match a query.

- AnchorEntity(.plane(...)) lets RealityKit do the ARKit anchoring; also (raycastResult:), (world:), (anchor:).
- ARView (UIKit, iOS 13): owns an ARSession; cameraMode .ar/.nonAR. RealityView (SwiftUI; visionOS 1, **iOS 18**): content.camera = .spatialTracking (passthrough + tracking) or .virtual; SpatialTrackingSession picks tracking and scene understanding (occlusion, shadows).
- Assets are **USD/USDZ**, authored in **Reality Composer Pro** (ships with Xcode). **SceneKit is deprecated** (WWDC25, soft: no removal date); ARSCNView deprecated in iOS 26 → RealityView.

## Example — a custom component + system (iOS 18)

```

struct Spin: Component { var speed: Float = 1 }
struct SpinSystem: System {
    static let q = EntityQuery(where: .has(Spin.self))
    init(scene: Scene) {}
    func update(context: SceneUpdateContext) {
        for e in context.entities(matching: Self.q,
            updatingSystemWhen: .rendering) {
            guard let s = e.components[Spin.self] else { continue }
            let dt = Float(context.deltaTime) // seconds
            e.transform.rotation = simd_quatf(angle: s.speed * dt,
                axis: [0, 1, 0]) * e.transform.rotation) } }
// once: Spin.registerComponent(); SpinSystem.registerSystem()
RealityView { content in
    content.camera = .spatialTracking // passthrough + tracking
    let a = AnchorEntity(.plane(.horizontal, classification: .any,
        minimumBounds: [0.2, 0.2]))
    let box = ModelEntity(mesh: .generateBox(size: 0.1),
        materials: [SimpleMaterial()])
    box.components.set(Spin(speed: 2)); a.addChild(box); content.add(a) }
  
```

## Traps

- Holding ARFrames** (a property, an async closure, slow Vision work on the delegate queue) drains the camera's buffer pool: "delegate is retaining N ARFrames", then the feed freezes. Copy out what you need.
- Placing content while .limited or before a plane settles → drift; planes grow and merge.
- No AR in the Simulator**; gate every feature on isSupported / supports... (LiDAR, face, geo). arkit in required capabilities only if AR is the whole app.
- No NSCameraUsageDescription → crash; face tracking also needs a privacy policy. Wrong reference-image **physical size** → content floats or sinks.
- Thermal:** camera + VIO + mesh + occlusion + rendering. Stop plane detection when done; pause() off screen.

## Remember

**"Camera + IMU → pose → anchors; entities = ids, components = data, systems = behaviour."**